

Supporting Information
For
Uranium(VI) reduction by iron(II) monosulfide mackinawite

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X-ray diffraction measurement

X-ray diffraction (XRD) samples were prepared by dropping 750 μL of mackinawite suspensions on glass slides and drying them in a vacuumed deccicator in the anaerobic glove box (4% H_2 with balance N_2). The prepared oriented samples were transferred in the vacuum deccicator to the X-ray diffractometer to minimize air exposure time. The X-ray diffraction patterns were measured with $\text{Cu } K\alpha$ radiation using a Rigaku Rotating Anode X-ray diffractometer (University of Michigan X-ray Microbeam Analysis Laboratory) operated at 40 kV and 100 mA. The patterns were measured at the scan speed of 2 degrees 2 theta per minute in a continuous scan mode.

Oxidation experiments

Oxidation experiments of mackinawite bound U(IV) were carried out using sealed glass batch reactors. Glass bottles (75 mL) tightly sealed with rubber septa and aluminum lids containing 5×10^{-5} mol/L uraninite reduced by 5 g/L mackinawite in 50 mL 0.1 M NaCl background electrolyte solution were used for the oxidation experiments. 1 mL (8.6×10^{-6} mole O_2 compared with 2.5×10^{-6} mole uraninite and 2.9×10^{-3} mole mackinawite per reactor) or 5 mL (4.3×10^{-5} mole O_2) air was injected using a gas-tight Hamilton syringe and needle through rubber septa into the reactors. Same amount of headspace gas was drawn out before injection of air to keep the headspace gas pressure constant. The air injected bottles were allowed to react for two days while continuously mixing the bottles with a shaker. After the two days of oxidation, the final pH was measured and solution samples and solid pastes were separated by centrifugation at 10,000

rpm. The solution samples were analyzed for dissolved U concentration by ICP-MS and the solid pastes were crimp sealed in glass septa vials, quenched at -80 °C, and shipped to SSRL for XAS data collection.

X-ray diffraction results

X-ray diffraction results are presented in Figure S3. At pH 5, the solid samples both with and without U(VI) addition have a pronounced (001) reflection of mackinawite at around two theta value of 17 degrees. The addition of U(VI) did not visibly change the XRD pattern. Other reflections of mackinawite are relatively weakened. There are sharp, strong reflections at around 32 and 45 degrees with less clear features at 27, 54, and 56 degrees 2 theta belonging to halite precipitated during the XRD sample preparation by vacuum drying of the mackinawite suspensions in the glove box. The origin of the prominent (001) reflection with weakened other peaks of the mackinawite at pH 5 and 7 is interpreted to be due to preferential, anisotropic dissolution of mackinawite particles along the *c*-axis at acidic and circum-neutral pH. The resulting thinning of the particles leads to a stronger tendency of preferred orientation along the *c*-axis, resulting in the pronounced (001) reflection. However, at the pH of 10, the relative intensities of all the reflections are similar to the original mackinawite sample. This pH is close to the mackinawite precipitation pH. Therefore, it can be deduced that dissolution of mackinawite is not as important at this pH. Contrary to the EXAFS analysis results, no noticeable uraninite feature is observed in the XRD patterns under all the pH conditions. This is due to the insufficient amounts of uraninite, given that the U loading was only 0.24 wt. %. The XRD observations suggest possibility of different U(VI) reaction mechanisms with mackinawite under different pH conditions, i.e., acidic and circum-neutral vs. basic conditions. Under acidic

and circum-neutral pH conditions, dissolution of mackinawite is an important process to consider in the uraninite formation by U(VI) reduction by mackinawite.

Results of U-reacted mackinawite oxidation

No detectable uranium release was observed from the oxygen injected reaction bottles, suggesting that mackinawite slows uraninite oxidation under the experimental conditions. Alternative explanation is that U is oxidized to U(VI) but is adsorbed by mackinawite or its oxidation products so that it is not released to solution. To test the two possibilities, EXAFS spectroscopy was used. EXAFS analysis results of U reacted with mackinawite at pH 7 after the 1 mL and 5 mL air injection are given Table S1 and Figure S3, along with model compounds uraninite and U(VI) solution. The EXAFS structural parameters of U from the air injected bottles were very similar to those before air injection (Table 1). This results confirm that U(IV) oxidation was inhibited in the presence of mackinawite under the experimental conditions.

References

- 1 Schofield, E.J.; Veeramani, H.; Sharp, J.O.; Suvorova, E.; Bernier-Latmani, R.; Mehta, A.; Stahlman, J.; Webb, S.M.; Clark, D.L.; Conradson, S.D.; Iltton, E.S.; Bargar, J.R. Structure of biogenic uraninite produced by *Shewanella oneidensis* strain MR-1. *Environ. Sci. Technol.* **2008**, *42*, 7898-7904.

Table S1. Structural parameters extracted from the EXAFS analysis of samples used in the oxidation experiments (*CN*: coordination number of backscatterer atoms around the absorber U atom, *R*: interatomic distance in Å unit of the absorber-backscatterer pair, σ^2 : Debye-Waller factor of the absorber-backscatterer pair, the subscript stands for the kind of backscatterer).

	Sample	U_SO_1	U_SO_2	U_Stock	Uraninite	Uraninite	Uraninite	UO _{2.00(s)}	Uraninite
Shell	Descripti on	pH = 7 Air 1mL	pH = 7 Air 5 mL	1000 ppm uranyl solution	Feff 8.10 calculated	Synthetic (this study)	Biogenic (this study)	Synthetic (Schofield et al., 2008) ¹	Biogenic (Schofield et al., 2008) ¹
O	<i>CN</i>	6.4(4)	7.9(1)		8	6.5(5)	7.2(8)	8	8
	<i>R</i>	2.34(5)	2.35(0)		2.3678	2.25(5)	2.33(4)	2.354(7)	2.347(7)
	σ^2	0.0072	0.0087		0	0.0164	0.0122	0.0046(7)	0.0103(6)
U ^{IV}	<i>CN</i>	7.7(2)	4.8(7)		12	2.5(2)	4.8(8)	12	6(1)
	<i>R</i>	3.85(0)	3.86(2)		3.8666	3.84(4)	3.83(7)	3.867(4)	3.846(5)
	σ^2	0.0070	0.0035		0	0.0071	0.0105	0.0029	0.0055(7)
O _{ax}	<i>CN</i>			2.4(5)					
	<i>R</i>			1.77(0)					
	σ^2			0.0030					
O _{eq}	<i>CN</i>			4.8(3)					
	<i>R</i>			2.42(3)					
	σ^2			0.0061					

Figure Captions

Figure S1. The 1st derivative U L_{III} edge XANES spectra of U-loaded mackinawite under different pH conditions, along with those of model compounds uraninite and U(VI) solution. The inset is the XANES and least squares fitting result of U(VI)-reacted mackinawite sample at pH 7 (line: measured data, dots: fit using spectra of uraninite and U(VI) model compound), showing U reduction to U(IV) upon reaction with mackinawite.

Figure S2. X-ray diffraction patterns of U(VI)-reacted mackinawite samples under different pH conditions.

Figure S3. U L_{III} edge EXAFS spectra (left panel) and corresponding Fourier transform magnitude functions (right panel) of U(VI) reacted with mackinawite at pH 7 measured after oxygen injection, along with those of model compound U(VI) aqueous solution and uraninite. A: Uranyl solution, B: 1 mL (9.4×10^{-6} mol O₂ per bottle vs 2.5×10^{-6} mol U per bottle) air injected, C: 5 mL air (4.7×10^{-5} mol O₂ per bottle) injected, and D: uraninite model compound (Solid lines: experimental data, Dots: fitting results).

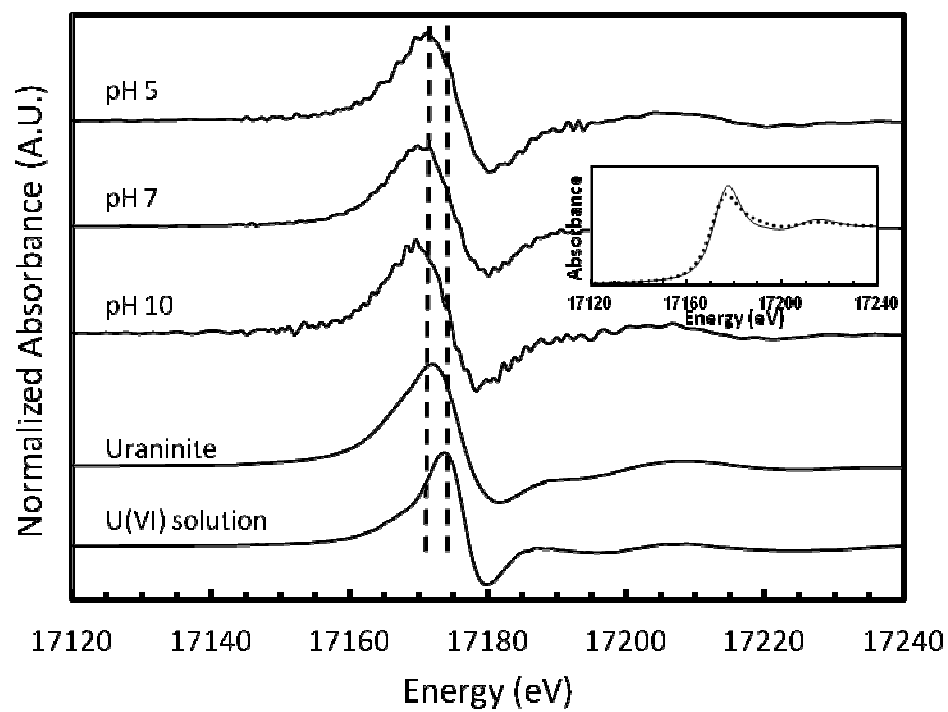


Figure S1

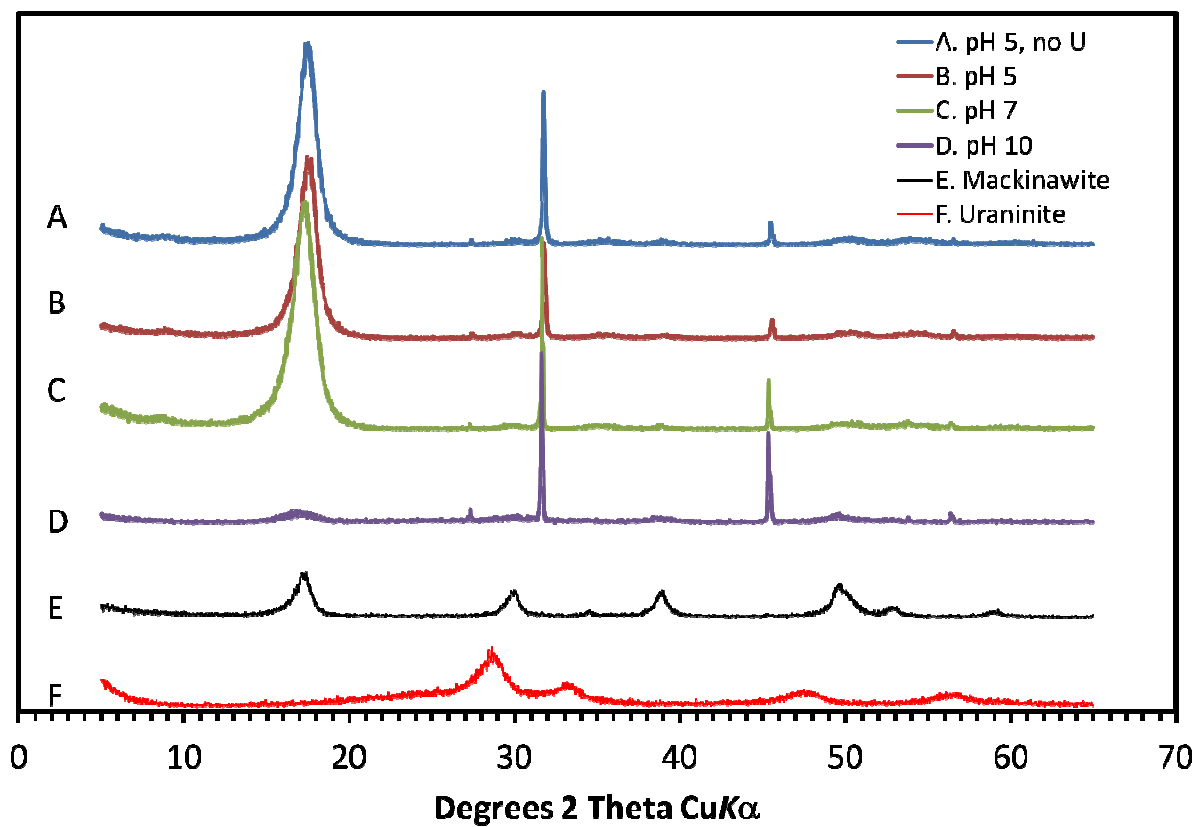


Figure S2

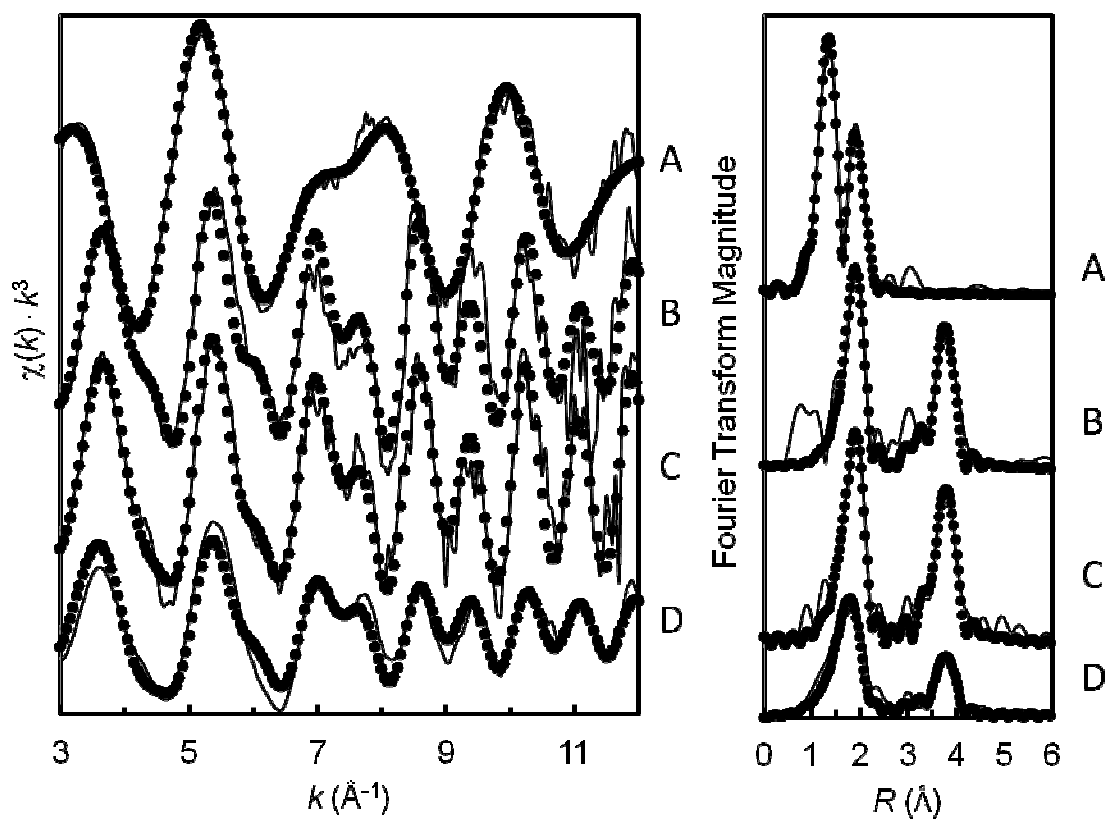


Figure S3